

# SIMATS ENGINEERING ASSESSMENT TOOLS I

COURSE NAME: Theory of Computation

COURSE CODE:

CSA1301

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## COURSE OUTCOME COVERED

**CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)**

*Assessment Tool Weightage: 10%*

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**QUESTIONS: 10\*2=20**

**Total Marks:20**

## Data Interpretation

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### Code of Conduct:

I Deepak 19241021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak

# SIMATS ENGINEERING ASSESSMENT TOOLS

Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language?

- ☒ A) ab
- ☐ B) aabb
- ☐ C) aaabbb
- ☐ D) abb

A, B, C

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

- ☒ A)  $a^n b^n$
- ☐ B)  $(ab)^n$
- ☐ C)  $a^n b^m$
- ☐ D)  $(a+b)^*$

A

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

- ☒ A)  $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$
- ☐ B)  $S \rightarrow ab \rightarrow aabb$
- ☐ C)  $S \rightarrow aSb \rightarrow ab$
- ☐ D)  $S \rightarrow aaSbb \rightarrow aabb$

A

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

- ☐ A) ab
- ☐ B) S
- ☒ C)  $\epsilon$
- ☐ D) aSb

C

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

- ☐ A) 2
- ☐ B) 3
- ☒ C) 4
- ☐ D) 5

C

~~Answer~~

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Consider:

$S \rightarrow SS \mid a$

## Q6. Ambiguity Check

This grammar is:

- A) Unambiguous
- ☒ B) Ambiguous
- C) Regular
- D) Deterministic

B

## Q7. String Generation

Which string can be generated?

- A) a
- B) aa
- C) aaa
- ☒ D) All of the above

D

## Q8. Interpretation of Ambiguity

Ambiguity means:

- A) One string  $\rightarrow$  one parse tree
- ☒ B) One string  $\rightarrow$  multiple parse trees
- C) No derivation
- D) Infinite symbols

B

## Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

## Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011?

- ☒ A)  $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$
- B)  $S \rightarrow 01 \rightarrow 0011$
- C)  $S \rightarrow 0S1 \rightarrow 01$
- D)  $S \rightarrow 00S11 \rightarrow 0011$

A

## Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

## Q10. Structural Analysis

Which of the following strings can be generated by the grammar?

- A) ab
- B) aab
- C) abab
- ☒ D) All of the Above

D

# SIMATS ENGINEERING ASSESSMENT TOOLS

## RUBRICS FOR CALCULATION

| Criteria                 | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|--------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| Conceptual Understanding | 30        | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| Accuracy of Answer       | 25        | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| Application of Knowledge | 20        | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| Completeness of Response | 15        | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| Clarity & Presentation   | 10        | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

82/100

*[Signature]*